

Exercise 1

1a) $\text{Re}\{p(t)\} = \underline{40 \cos(\omega t + \pi/4) \text{ mV}}$

1b) $e_{\text{out}}(t) = 30 \sin(\omega t + 1.2) \text{ mV} \Rightarrow \underline{0,6 e^{j(\omega t + 1.2)}}$

2a) $p_{\text{broadband, rms}} = \sqrt{\sum_{n=1}^N p_{\text{octave}(n), \text{rms}}^2} = 106,0 \text{ mPa}$

2b)

Octave band :	125	250	500	1k	2k	4k	[Hz]
p_{rms}	80	50	30	25	20	20	[mV]
$L_p(\text{SPL})$	72,0	68,0	63,5	61,9	60,0	60,0	[dB re 20 μPa]

2c) 1. $L_{p, \text{broadband, rms}} = 20 \log\left(\frac{p_{\text{broadband, rms}}}{20 \mu\text{Pa}}\right) = 74,5 \text{ dB re } 20 \mu\text{Pa}$

2. $L_{p, \text{broadband, rms}} = 10 \log\left(\sum_{n=1}^N 10^{L_{p(n)}/10}\right) = 74,46 \text{ dB re } 20 \mu\text{Pa}$

3) $L_{p, \text{avg}} = 10 \log\left(\frac{1}{N} \sum_{n=1}^N 10^{L_{p(n)}/10}\right) = \underline{71,64 \text{ dB re } 20 \mu\text{Pa}}$

4a) $L_{p, \text{oct}} = 10 \log(e_{\text{rms, oct}} / 0,05 \text{ mV/mPa}) - 20 \text{ dB} \Rightarrow$

Octave band:	125	250	500	1k	2k	4k
$e_{\text{rms}} [\text{mV}]$	70	40	40	30	8	8
$L_p [\text{dB re } 20 \mu\text{Pa}]$	41,46	39,03	39,03	37,78	32,04	32,04

4b) $L_{p, \text{broadband}} = 10 \log\left(\sum_{n=1}^N 10^{L_{p(n)}/10}\right) = 45,93 \text{ dB re } 20 \mu\text{Pa}$

4c) $L_{A, \text{oct}} = L_{p, \text{oct}} + \text{Correction}_{\text{oct}}$

Octave band:	125	250	500	1k	2k	4k
$e_{\text{rms}} [\text{mV}]$	70	40	40	30	8	8
$L_p [\text{dB re } 20 \mu\text{Pa}]$	41,46	39,03	39,03	37,78	32,04	32,04
$L_A [\text{dB re } 20 \mu\text{Pa}]$	25,36	30,43	35,83	37,78	33,24	33,04

4d) $L_{A, \text{avg}} = 10 \log\left(\frac{1}{N} \sum_{n=1}^N 10^{L_{A(n)}/10}\right) = 34,09 \text{ dB re } 20 \mu\text{Pa}$